

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Case Study

Code of Conduct:

I Balakumaran M (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.
2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.
3. A retail company wants to segment its customers based on purchasing behavior, but the data contains hidden patterns and incomplete information. Apply the EM algorithm to identify clusters and estimate underlying distributions. Analyze convergence behavior, initialization challenges, and the role of probability learning. Discuss how the results can be used for targeted marketing and evaluate limitations of the approach.
4. A research team is developing a concept learning system to classify medical images into normal and abnormal categories. Using hypothesis space search and the Candidate Elimination approach, design a model that learns from labelled examples. Analyze the impact of finite vs infinite hypothesis spaces, sample complexity, and mistake bound model on learning performance. Evaluate how well the system generalizes to unseen data.
5. A financial institution needs to make investment decisions under uncertain market conditions. Using Bayesian inference, Gibbs algorithm, and probability learning, design a system to predict outcomes and update beliefs dynamically. Analyze how prior knowledge influences predictions, evaluate risk using posterior probabilities, and discuss how the model adapts as new data becomes available. Consider computational challenges and optimization strategies.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q1. Intelligent Disease Diagnosis using Bayesian Methods

Given: Hospital diagnosis system using patient symptoms, medical history and probabilistic learning.

Model Design: Use Bayes' Theorem to compute disease probability, Naïve Bayes for fast classification, and a Bayesian Belief Network (BBN) to model dependencies among symptoms such as fever, cough and chest pain.

Analysis: Bayes' theorem updates disease probability as new symptoms arrive. Naïve Bayes assumes conditional independence, while BBN captures symptom relationships, improving diagnosis when symptoms influence one another.

Real-world Constraints: Missing patient data, noisy records and changing disease prevalence reduce accuracy. Prior probabilities should be updated periodically using recent hospital data.

Conclusion: A combined Bayesian model provides accurate, explainable diagnosis and effectively handles uncertainty through probability-based reasoning.

Q2. Spam Detection using Naïve Bayes

Given: Email classification into Spam or Ham.

System Design: Extract features such as keywords, sender information and links. Train a Naïve Bayes classifier. Bayes Optimal Classifier represents the ideal theoretical classifier.

MLE vs MDL: MLE selects parameters that maximize observed data likelihood. MDL balances model accuracy with simplicity, reducing overfitting by avoiding unnecessary features.

Performance Analysis: Larger training samples improve accuracy. Very large hypothesis spaces increase computation and overfitting risk, while appropriate feature selection improves generalization.

Conclusion: Naïve Bayes provides efficient spam detection and MDL helps build a simpler, more reliable model.

Q3. Customer Segmentation using EM Algorithm

Given: Retail customer purchasing data with hidden patterns and incomplete information.

Method: Use the Expectation-Maximization (EM) algorithm to estimate hidden cluster memberships and distribution parameters iteratively.

Analysis: Initialization affects convergence and EM may converge to a local optimum. Better initialization improves clustering quality.

Application: Clusters help identify premium, occasional and discount customers for personalized marketing campaigns.

Conclusion: EM effectively handles incomplete probabilistic data but requires careful initialization and validation.

Q4. Medical Image Classification using Candidate Elimination

Given: Labelled medical images classified as Normal or Abnormal.

Model Design: Represent hypotheses using image features and update the Specific (S) and General (G) boundaries with Candidate Elimination.

Analysis: Finite hypothesis spaces converge faster, while infinite spaces require more samples. The mistake bound decreases as informative training examples increase.

Generalization: Balanced datasets and representative features improve prediction on unseen images.

Conclusion: Candidate Elimination provides interpretable concept learning with effective generalization when sufficient labelled data is available.

Q5. Investment Prediction using Bayesian Inference

Given: Financial investment decisions under uncertain market conditions.

System Design: Use Bayesian inference to update market beliefs. Gibbs sampling approximates posterior distributions when exact inference is difficult.

Analysis: Prior knowledge influences early predictions, while posterior probabilities increasingly reflect observed market data.

Challenges: High-dimensional data increases computation. Efficient sampling and parallel processing improve performance.

Conclusion: Bayesian learning enables dynamic investment prediction by continuously updating probabilities as new market information becomes available.